Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M2:Practical – VIII

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Class: TYIT	Batch: 1
Date of Assignment: 30/09/2026	Date/Time of Submission: 30/09/2026

8.API Creation and Integration

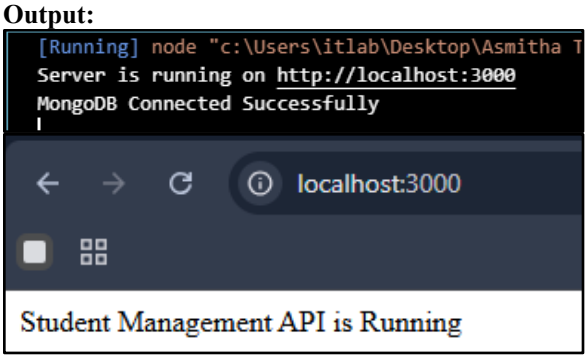
a. Create a simple REST API using Express

Code:

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
```

Output:



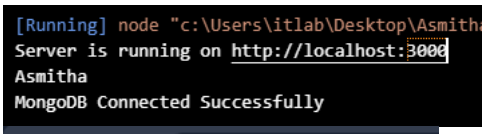
b. Implement a GET API to retrieve data.

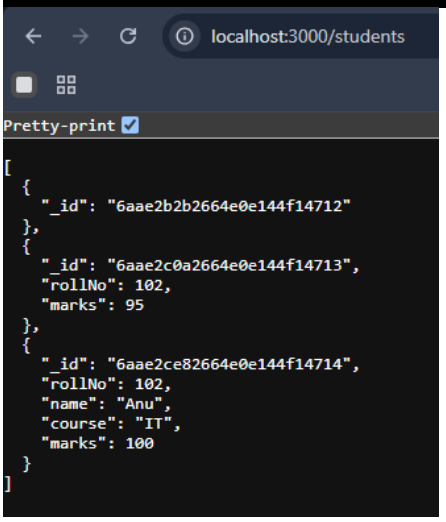
Code:

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output:





c. Fetch data from the API in a React application.

Code:

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Simple REST API
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  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

App.jsx:

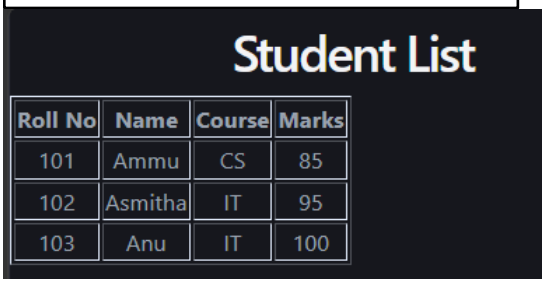
```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => {
    axios.get("http://localhost:3000/students")
    .then((response) => { setStudents(response.data); })

    .catch((error) => { console.log(error); });
  }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
export default App;
```



Output:





d. Send data from React to the server using a POST request.
e. Store data in MongoDB.
f. Display stored data in React page.
Code:
server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("students")
      .insertOne(student);
    res.json({
      message: "Student added successfully",
      student: student
    });
  }
  catch (error) {
    res.status(500).json({
      message: "Error adding student"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on
http://localhost:${PORT}`);
});
```

App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
    .then((response) => { setStudents(response.data); })
    .catch((error) => { console.log(error); });
  };
  useEffect(() => { getStudents(); }, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios.post("http://localhost:3000/students",
newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
```

```
            setName("");
            setCourse("");
            setMarks("");
            getStudents();
          })
          .catch((error) => {
            console.log(error);
          });
    };
    return (
      <div>
        <h1>Student Management System</h1>
        <h2>Add Student</h2>
        <input
          type="number"
          placeholder="Roll No"
          value={rollNo}
          onChange={(e) => setRollNo(e.target.value)}
        />
        <br /><br />
        <input
          type="text"
          placeholder="Student Name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />
        <br /><br />

        <input
          type="text"
          placeholder="Course"
          value={course}
          onChange={(e) => setCourse(e.target.value)}
        />
        <br /><br />
        <input
          type="number"
          placeholder="Marks"
          value={marks}
          onChange={(e) => setMarks(e.target.value)}
        />
        <br /><br />
        <button onClick={addStudent}>
          Add Student
        </button>
        <hr />
        <h2>Student List</h2>
        <table border="1">
          <thead>
            <tr>
              <th>Roll No</th>
              <th>Name</th>
              <th>Course</th>
              <th>Marks</th>
            </tr>
          </thead>
          <tbody>
            {students.map((student) => (
              <tr key={student._id}>
                <td>{student.rollNo}</td>
                <td>{student.name}</td>
                <td>{student.course}</td>
                <td>{student.marks}</td>
              </tr>
            ))}
          </tbody>
        </table>

      </div>
    );
}
export default App;
```

Output:

